**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 106 · Lesson 106 · What an Operating System Does

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Distinguish RAM from virtual storage.

Q2. In a BST, where is a value less than the current node placed?

Q3. Why can an array provide direct indexed access?

Q4. A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

operating system — System software managing hardware resources and services.

process — An executing instance of a program.

device driver — Software allowing the OS to communicate with hardware.

spooling — Queuing data such as print jobs for a slower peripheral.

## New learning

Manage processes — The OS creates and schedules processes; programs share CPU time.

Manage resources — Memory, files and device access are mediated by the OS.

Provide services — Interfaces, user accounts and access controls make a computer usable.

## Worked example

A word processor saves a document while music is playing. Explain three OS services involved.

1. Model answer — The OS allocates CPU time to both processes; the file system writes the document to storage; a device driver controls sound output. Memory management keeps process data protected. — Check: link each claim to the scenario and use precise OCR terminology.

# We Do · Practise with guidance

A pupil prints while a browser downloads a file. Which OS responsibilities are involved?

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

Identify concurrent processes; explain the print queue/device driver; describe file and network access. Printing and downloading share resources without either application controlling the hardware directly.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Describe three functions of an operating system, linking each to a user action. [6]

### Question 2

A computer has three applications requesting the printer. Explain how the OS can handle this. [6]

### Question 3

Evaluate the claim that application programs can directly manage all hardware without system software. [8]

# Exit Ticket · Answer before checking

The operating system only provides a graphical user interface.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. RAM is local working memory; virtual storage presents remote/pooled file storage.

Q2. In its left subtree (under the chosen no-duplicate convention).

Q3. Its element locations can be calculated from base address and index.

Q4. Sum 0 and carry-out 1.

### You Do pages · Independent questions

Q1. Up to 2 marks for each explained service: process scheduling, memory management, file management, device management, security or interface.

Q2. It manages a print queue/spooling [2], coordinates each print job in turn [2], and uses a driver to communicate with the printer [2].

Q3. Identify need for abstraction and drivers [2], arbitration of shared hardware [2], memory/process protection [2], and a reasoned conclusion with example [2].

### Exit Ticket · Model answer

False. It also manages processes, memory, files, devices and security.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding